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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
First Semester M.Tech. (EVT)(CBCS) Degree Examinations
January 2023

22EVT111 - ADVANCED BATTERY TECHNOLOGY FOR ELECTRIC VEHICLES

Duration: 3 Hours

Max. Marks:100

*Note: Answer **Five full questions** choosing **Two full questions** from **Unit – I & Unit – II each** and **One full question** from **Unit – III**.*

Unit – I

	Marks	BT*	CO*	PO*
1. a) Explain the design of flooded lead-acid battery.	7	L*2	1	3
b) Brief out the chemical functioning and reaction of fuel cell.	4	L2	1	3
c) Highlight the effects of Temperature on VRLA and NiMH battery formation.	9	L3	2	3
2. a) What are the salient features of the traction battery for EVs.	7	L2	1	3
b) List the characteristics of the Li-ion battery that are favorable for EV applications.	4	L2	1	3
c) Discuss the criteria that are useful to determine the end of formation of a VRLA battery.	9	L3	2	3
3. a) Enumerate the four steps of the conventional processing method for manufacturing the AB5 battery materials.	8	L2	1	3
b) Discuss the storage of VRLA and NIMH batteries.	6	L3	2	3
c) Explain the effects of excessive heat on battery cycle life.	6	L3	2	3

Unit – II

4. a) Explain the charging of NIMH batteries with the relevant chemical reactions that takes place during charging process.	4	L2	3	3
b) What are the prerequisites for fast-charging?	7	L2	3	3
c) Explain the combined effect of each of the factors affecting the discharge of a traction battery.	9	L3	4	3
5. a) Explain the AC impedance and AC conductance tests that are performed to determine battery pack internal impedance of the traction battery.	7	L2	3	3
b) Brief out conductive coupling and inductive coupling.	4	L2	3	3
c) List a set of initial conditions to successfully perform range tests on the EV.	9	L3	4	3
6. a) Explain the three primary methods of charging EV batteries.	8	L2	3	3
b) What are the observations made by the cold weather performance on a compact size EV using VRLA batteries?	6	L3	4	3
c) What is the necessity of performing dynamic stress test (DST) for EV battery packs?	6	L3	4	3

Unit – III

7. a) Explain the BPMS charge protector and charge indicator.	8	L2	5	3
b) Explain the depolarization process to enhance the charging.	6	L3	5	3
c) Explain how a smart battery removes the charge control from the battery charger and assigns it to the battery.	6	L4	5	3
8. a) Explain in detail the Model of BPMS.	7	L2	5	3
b) Explain the BPMS thermal management system.	7	L3	5	3
c) What are the important issues that must be addressed in understanding the types of traction batteries.	6	L4	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
